Type MS · Connected rates: linking two changing quantities

Answer 1 of 7 · Question 9 part whole · 7 marks

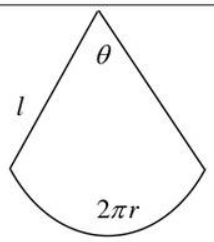
2023 JAN · Paper 02 · find part whole in this mark scheme extract

9		
	$\frac{dA}{dt} = 0.45$	B1
	$V = x^3 \Rightarrow \frac{dV}{dx} = 3x^2$	B1
	$A = 6x^2 \Rightarrow \frac{dA}{dx} = 12x$	B1
	$384 = 6x^2 \Rightarrow x = 8$	M1
	$\frac{dV}{dt} = \frac{dV}{dx} \times \frac{dx}{dA} \times \frac{dA}{dt}$	M1
	$\frac{dV}{dt} = 3x^2 \times \frac{1}{12x} \times 0.45 \left[= \frac{9x}{80} \right]$ oe	dM1
	When $x = 8$ $\frac{dV}{dt} = 0.9\,\text{cm}^3/\text{s}$	A1
		Total 7 marks

Type MS · Connected rates: linking two changing quantities

Answer 2 of 7 · Question 10 part c · 5 marks

2020 NOV · Paper 01 · extract 1 of 3 · find part c in this mark scheme extract

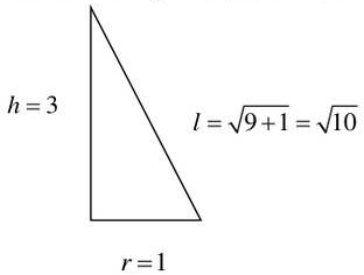
Number		
10(a)	Circumference of base $= 2\pi r$ 	B1
	$l\theta = 2\pi r \Rightarrow \theta = \dfrac{2\pi r}{l}$	B1
	$A = \dfrac{1}{2}l^2\theta = \dfrac{1}{2}l^2\dfrac{2\pi r}{l} = \pi rl$	M1A1 (4)
(b)	$A = \pi rl$	
	$l = r\sqrt{10} \Rightarrow A = \pi r^2\sqrt{10}$	B1
	$\dfrac{dA}{dr} = 2\pi r\sqrt{10} \Rightarrow k = 2\sqrt{10}$	M1A1 (3)
(c)	$\dfrac{dV}{dt} = 1.5\ \left(\text{cm}^3/\text{s}\right)$	B1
	$\dfrac{dA}{dt} = \dfrac{dA}{dr} \times \dfrac{dr}{dV} \times \dfrac{dV}{dt}$	M1
	$V = \dfrac{1}{3}\pi r^2 h = \dfrac{1}{3}\pi r^2 \times 3r = \pi r^3$	B1
	$\dfrac{dV}{dr} = 3\pi r^2$	B1ft
	$\therefore \dfrac{dA}{dt} = 2\pi \times 8\sqrt{10} \times 1.5 \times \dfrac{1}{3\times 64\pi} = 0.3952... = 0.395\ \text{cm}^2/\text{s}$	A1 (5)
		[12]

Part	Mark	Notes	
(a)	B1	For the circ of the base $L = 2\pi r$	
	B1	$R = l$ and $l = r\theta$ Therefore $l\theta = 2\pi r \Rightarrow \theta = \dfrac{2\pi r}{l}$	$R = l$
	M1	$A = \dfrac{1}{2}l^2\theta$ and substituting their expression for θ to give $A = \dfrac{1}{2}l^2\theta = \dfrac{1}{2}l^2\dfrac{2\pi r}{l}$	Uses the formula $A = \dfrac{1}{2}RL$ $A = \dfrac{1}{2}\times l \times 2\pi r \Rightarrow \left(A = \pi rl\right)$
	A1	For the required expression for A, $A = \pi rl$ with no errors.	

Type MS · Connected rates: linking two changing quantities

Answer 2 of 7 · Question 10 part c · 5 marks

2020 NOV · Paper 01 · extract 2 of 3 · find part c in this mark scheme extract

(b)	B1	For finding that the slant height is $\sqrt{10}$ times the radius of the cone  So $l=r\sqrt{10}$
	M1	Substitutes $l=r\sqrt{10}$ into the given expression $A=\pi rl$ and differentiates their resulting expression to find $\frac{dA}{dr}$ $A=\pi r^2\sqrt{10}$ so $\frac{dA}{dr}=2\pi r\sqrt{10}$
	A1	Therefore $k=2\sqrt{10}$
(c)	B1	States $\frac{dV}{dt}=1.5\ \left(\text{cm}^3/\text{s}\right)$ Award if it seen explicitly in (b) **and used** in (c)
	M1	States (or uses) a correct chain rule $\frac{dA}{dt}=\frac{dA}{dr}\times\frac{dr}{dV}\times\frac{dV}{dt}$
	B1	For finding the volume of a cone in terms of r only $V=\frac{1}{3}\pi r^2h=\frac{1}{3}\pi r^2\times 3r=\pi r^3$
	B1ft	Differentiates their expression for the volume of a cone provided it is in terms of V and r only. $\frac{dV}{dr}=3\pi r^2$
	A1	For combining all required terms into their chain rule and evaluating to 3 significant figures, $\frac{dA}{dt}=2\pi\times 8\sqrt{10}\times 1.5\times\frac{1}{3\times 64\pi}=0.3952...=0.395\ \text{cm}^2/\text{s}$
	ALT – in terms of h	
	B1	States $\frac{dV}{dt}=1.5\ \left(\text{cm}^3/\text{s}\right)$ Award if it seen explicitly in (b) **and used** in (c)
	M1	States (or uses) a correct chain rule $\frac{dA}{dt}=\frac{dA}{dh}\times\frac{dh}{dt}$ and $\frac{dV}{dt}=\frac{dV}{dh}\times\frac{dh}{dt}$
	B1	For finding the area and volume of a cone in terms of h $r=\frac{h}{3}$, $A=\pi r^2\sqrt{10}\Rightarrow A=\frac{\sqrt{10}}{9}\pi h^2$ and $V=\frac{1}{3}\pi r^2h\Rightarrow V=\frac{1}{27}\pi h^3$
	B1ft	Differentiates their expressions for the area and volume of a cone provided they are both in terms of h only. $\frac{dA}{dh}=\frac{2\sqrt{10}}{9}\pi h$ and $\frac{dV}{dh}=\frac{3}{27}\pi h^2$
	A1	Combines the required terms into their chain rules and evaluating to 3 significant figures $\frac{dA}{dt}=0.395\ 102$

Type MS · Connected rates: linking two changing quantities

Answer 2 of 7 · Question 10 part c · 5 marks

2020 NOV · Paper 01 · extract 3 of 3 · find part c in this mark scheme extract

Question Number	Scheme	Marks
	1	

Type MS · Connected rates: linking two changing quantities

Answer 3 of 7 · Question 6 part b · 8 marks

2023 JUN · Paper 01 (regional R) · extract 1 of 4 · find part b in this mark scheme extract

6(a)	$\tan 30^\circ = \frac{r}{h} \Rightarrow r = \frac{h}{\sqrt{3}}$	B1
	$V = \frac{1}{3}\pi r^2 h \Rightarrow V = \frac{1}{3}\pi\left(\frac{h}{\sqrt{3}}\right)^2 h \Rightarrow V = \frac{1}{9}\pi h^3$ *	M1A1 cso [3]
(b)	$\frac{dV}{dt} = -4$	B1
	$A = \pi r^2 \Rightarrow A = \pi\left(\frac{h}{\sqrt{3}}\right)^2 = \frac{\pi h^2}{3}$	B1
	$\frac{dA}{dh} = \frac{2\pi h}{3}$ $\quad \frac{dV}{dh} = \frac{3\pi h^2}{9} = \frac{\pi h^2}{3}$	M1M1A1
	$\frac{dA}{dt} = \frac{dA}{dh} \times \frac{dV}{dt} \times \frac{1}{\frac{dV}{dh}}$	M1
	$\frac{dA}{dt} = \frac{2\pi h}{3} \times (-4) \times \frac{3}{\pi h^2} = -\frac{8}{h} \Rightarrow \frac{dA}{dt} = -\frac{8}{24} = -\frac{1}{3}$ $\left(\text{cm}^2/\text{s}\right)$	dM1A1 [8]
	ALT – working in terms of *r* rather than *h*	
	$\frac{dV}{dt} = -4$	[B1
	$V = \frac{1}{3}\pi r^2 h \Rightarrow V = \frac{1}{3}\pi r^2\left(\sqrt{3}r\right) = \frac{\sqrt{3}}{3}\pi r^3$	B1
	$A = \pi r^2 \Rightarrow \frac{dA}{dr} = 2\pi r$ $\quad V = \frac{\sqrt{3}}{3}\pi r^3 \Rightarrow \frac{dV}{dr} = \sqrt{3}\pi r^2$	M1M1A1
	$\frac{dA}{dt} = \frac{dA}{dr} \times \frac{1}{\frac{dV}{dr}} \times \frac{dV}{dt}$	M1
	$\frac{dA}{dt} = 2\pi r \times \frac{1}{\sqrt{3}\pi r^2} \times -4 = -\frac{8}{\sqrt{3}r} \Rightarrow \frac{dA}{dt} = -\frac{8}{\sqrt{3}\times\frac{24}{\sqrt{3}}} = -\frac{1}{3}$	dM1A1]
		Total 11 marks

Type MS · Connected rates: linking two changing quantities

Answer 3 of 7 · Question 6 part b · 8 marks

2023 JUN · Paper 01 (regional R) · extract 2 of 4 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	B1	For $r = \frac{h}{\sqrt{3}}$ or $r^2 = \frac{1}{3}h^2$ $r = \frac{h}{\sqrt{3}}$ may be implied by substitution into the correct formula for volume of a cone.
	M1	For using the correct volume of a cone and substituting in their expression for r or r^2 to find an expression for the volume in terms of h
	A1	For the correct expression with no errors seen. Must have $V =$
(b)	B1	For stating $\frac{dV}{dt} = -4$ (allow +) May be embedded in working.
	B1	For finding the correct expression for the area of a circle in terms of h Note: if differentiation is attempted before writing in terms of h then award 1st M1 for $\frac{dA}{dh} = \frac{dA}{dr} \times \frac{dr}{dh}$ with attempts at $\frac{dA}{dr}, \frac{dr}{dh}$ found and substituted in and attempt to substitute for r in terms of h. B1 once $\frac{dA}{dh}$ found completely in terms of h
	M1	For attempting to differentiate their expression for A in terms of h See general guidance for what constitutes an attempt to differentiate. A must have the form $A = kh^2$ (dimensionally correct for area).
	M1	For attempting to differentiate the given V See general guidance for what constitutes an attempt to differentiate.
	A1	Both derivatives correct. $\frac{dA}{dh}$ and $\frac{dV}{dh}$
	M1	For a correct expression of chain rule to find $\frac{dA}{dt}$
	dM1	For substituting their derivatives into a correct chain rule. Dep on previous three M marks.
	A1	For the correct $\frac{dA}{dt} = -\frac{1}{3}$ $\left(\text{cm}^2/\text{s}\right)$ Allow $-0.\dot{3}$ but not -0.3. ISW once exact answer found.
	ALT	
	B1	For stating $\frac{dV}{dt} = -4$ (allow +) May be embedded in working.
	B1	For finding the correct expression for the volume in terms of r Must be working with correct volume expression as given in (a).
	M1	For attempting to differentiate expression for A in terms of r See general guidance for what constitutes an attempt to differentiate. A must have the form $A = kr^2$ (dimensionally correct for area).
	M1	For attempting to differentiate expression for V in terms of r See general guidance for what constitutes an attempt to differentiate.

Type MS · Connected rates: linking two changing quantities

Answer 3 of 7 · Question 6 part b · 8 marks

2023 JUN · Paper 01 (regional R) · extract 3 of 4 · find part b in this mark scheme extract

	A1	Both derivatives correct. $\frac{dA}{dr}$ and $\frac{dV}{dr}$
	M1	For a correct expression of chain rule to find $\frac{dA}{dt}$
	dM1	For substituting their derivatives into a correct chain rule. Dep on previous three M marks.
	A1	For the correct $\frac{dA}{dt} = -\frac{1}{3}$ $\left(\text{cm}^2/\text{s}\right)$ Allow $-0.\dot{3}$ but not -0.3. ISW once exact answer found.

Type MS · Connected rates: linking two changing quantities

Answer 3 of 7 · Question 6 part b · 8 marks

2023 JUN · Paper 01 (regional R) · extract 4 of 4 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · Connected rates: linking two changing quantities

Answer 4 of 7 · Question 9 part b · 8 marks

2025 NOV · Paper 01 · extract 1 of 4 · find part b in this mark scheme extract

Question	Scheme	Marks
9(a)	$V = 0.2t$	B1
	$r = h\tan 30 = \dfrac{h}{\sqrt{3}}$	B1
	$V = \dfrac{1}{3}\pi\left(\dfrac{h}{\sqrt{3}}\right)^2 h \quad \left(=\dfrac{\pi}{9}h^3\right)$	B1
	$0.2t = \dfrac{\pi}{9}h^3 \Rightarrow h = \sqrt[3]{\dfrac{9t}{5\pi}}$ *	M1A1 cso [5]
(b)	$\dfrac{dV}{dt} = 0.2$	B1
	$A = \pi r^2 = \dfrac{\pi h^2}{3}$	B1
	$\dfrac{dA}{dh} = \dfrac{2}{3}\pi h, \quad \dfrac{dV}{dh} = \dfrac{\pi h^2}{3}$	M1, M1
	when $t = 6$	
	$h = \sqrt[3]{\dfrac{9\times 6}{5\pi}} \quad (=1.50923...)$	B1
	$\dfrac{dA}{dt} = \dfrac{dV}{dt}\times\dfrac{1}{\dfrac{dV}{dh}}\times\dfrac{dA}{dh} = \left[\dfrac{1}{5}\times\dfrac{3}{\pi h^2}\times\dfrac{2\pi h}{3}\right]$	M1
	$\dfrac{dA}{dt} = \dfrac{2}{5h} = \dfrac{2}{5\times 1.50923} \approx 0.265$	M1A1 [8]
	ALT	
	$\dfrac{dV}{dt} = 0.2$	[B1]
	$A = \pi r^2 = \dfrac{\pi h^2}{3}$	[B1]
	$\dfrac{dA}{dh} = \dfrac{2}{3}\pi h$	[M1]
	$h^3 = \dfrac{9t}{5\pi} \Rightarrow t = \dfrac{5}{9}\pi h^3$	
	$\dfrac{dt}{dh} = \dfrac{5}{3}\pi h^2$	[M1]
	When $t = 6$	
	$h = \sqrt[3]{\dfrac{9\times 6}{5\pi}} \quad (=1.50923..).$	[B1]
	$\dfrac{dA}{dt} = \dfrac{dA}{dh}\times\dfrac{dh}{dt}$	

Type MS · Connected rates: linking two changing quantities

Answer 4 of 7 · Question 9 part b · 8 marks

2025 NOV · Paper 01 · extract 2 of 4 · find part b in this mark scheme extract

	$= \frac{2}{3}\pi h \times \frac{1}{\frac{5}{2}\pi h^3} = \frac{2}{5h}$ $\frac{dA}{dt} = \frac{2}{5\times 1.50923} \approx 0.265$	[M1] [M1A1] [8]
		Total 13 marks

Type MS · Connected rates: linking two changing quantities

Answer 4 of 7 · Question 9 part b · 8 marks

2025 NOV · Paper 01 · extract 3 of 4 · find part b in this mark scheme extract

Part	Mark	Notes
9 (a)	B1	For stating $V = 0.2t$
	B1	For finding h in terms of r
	B1	For finding the volume of the cone in terms of h only
	M1	Equates both expressions for V
	A1 cso	For obtaining the required expression
(b)	B1	For stating $\frac{dV}{dt} = 0.2$ anywhere in their response, including part (a).
	B1	For finding the area of the surface of liquid in terms of h only.
	M1	For the correct $\frac{dA}{dh}$
	M1	For the correct $\frac{dV}{dh}$
	B1	For using the correct expression for h at $t = 6$
	M1	For a correct expression of chain rule in terms of V, A, h and t only
	M1	For substituting all the values
	A1	For the correct value of awrt 0.265
ALT (b)		
	B1	For stating $\frac{dV}{dt} = 0.2$
	B1	For finding the area of the surface of liquid in terms of h only.
	M1	For the correct $\frac{dA}{dh}$
	M1	For rearranging the expression in (a) and finding $\frac{dt}{dh}$
	B1	For using the correct expression for h at $t = 6$
	M1	For a correct expression of chain rule in terms of V, A, h and t only
	M1	For substituting all the values
	A1	For the correct value of awrt 0.265

Type MS · Connected rates: linking two changing quantities

Answer 4 of 7 · Question 9 part b · 8 marks

2025 NOV · Paper 01 · extract 4 of 4 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · Connected rates: linking two changing quantities

Answer 5 of 7 · Question 9 part b · 3 marks

2021 JUN · Paper 02 · extract 1 of 3 · find part b in this mark scheme extract

9 (a)	$\frac{dA}{dt} = 0.03$	B1
	$A = \frac{1}{2}x^2 \sin 60^\circ = \frac{\sqrt{3}}{4}x^2$	M1
	$\frac{dA}{dx} = \frac{\sqrt{3}}{2}x$	A1
	When $x = 2$ $\frac{dx}{dt} = \frac{1}{\sqrt{3}} \times 0.03 = 0.0173$ cm/s	M1 A1 (5)
(b)	$V = \sqrt{3}x^3$ $\frac{dV}{dx} = 3\sqrt{3}x^2$	M1
	When $x = 2$ $\frac{dV}{dt} = 12\sqrt{3} \times 0.0173 = 0.36$	M1 A1 (3)
		Total 8 marks

Type MS · Connected rates: linking two changing quantities

Answer 5 of 7 · Question 9 part b · 3 marks

2021 JUN · Paper 02 · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Guidance
(a)	B1	For stating or using correctly in their Chain Rule $\frac{dA}{dt}=0.03$
	M1	For using the **correct** formula $\left(\frac{1}{2}ab\sin C\right)$ with the correct lengths and angle of 60° or $\frac{\pi}{3}$, for the cross-sectional area of the prism to obtain $A=\frac{1}{2}x^2\sin 60^\circ=\left(\frac{\sqrt{3}}{4}x^2\right)$ and differentiating their expression which must be as a minimum $A=px^2$ to obtain $\frac{dA}{dx}=qx$ [where p and q are constants]. [The height of the triangle is $\frac{\sqrt{3}}{2}x$ if they use $\frac{1}{2}\times\text{base}\times\text{height}$]
	A1	For the correct $\frac{dA}{dx}=\frac{\sqrt{3}}{2}x$
	M1	For applying a **correct** Chain rule using their $\frac{dA}{dx}$ and $x=2$ to obtain $\frac{dx}{dt}=\left(\frac{1}{\frac{dA}{dx}}\times\frac{dA}{dt}\right)=\frac{dx}{dA}\times\frac{dA}{dt}=\frac{2}{\sqrt{3}}\times\frac{1}{2}\times 0.03$
	A1	$\frac{dx}{dt}=0.0173$
(b)	M1	For a correct expression for the volume using their A from part (a) to obtain $V=\frac{\sqrt{3}}{4}x^2\times 4x=\left(\sqrt{3}x^3\right)$ and differentiating their expression which must be as a minimum $V=mx^3$ to obtain as a minimum $\frac{dV}{dx}=nx^2$ [where m and n are constants] $\left(\frac{dV}{dx}=3\sqrt{3}x^2\right)$
	M1	For applying a correct Chain rule using their $\frac{dV}{dx}$ and $x=2$ to obtain $\frac{dV}{dt}=\frac{dV}{dx}\times\frac{dx}{dt}=12\sqrt{3}\times 0.0173=\lceil 0.359...\rceil$ $\left(\text{ft their }\frac{dx}{dt}\right)$ **Note:** $\frac{dx}{dt}=0.0173$ or $\frac{\sqrt{3}}{100}$ $\left(\frac{dV}{dt}=\frac{dV}{dx}\times\frac{dx}{dt}=12\sqrt{3}\times\frac{\sqrt{3}}{100}=\frac{9}{25}=0.36\right)$
	A1	For awrt 0.36

Type MS · Connected rates: linking two changing quantities

Answer 5 of 7 · Question 9 part b · 3 marks

2021 JUN · Paper 02 · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Connected rates: linking two changing quantities

Answer 6 of 7 · Question 10 part whole · 9 marks

2020 JAN · Paper 02 · extract 1 of 2 · find part whole in this mark scheme extract

10	$\frac{dV}{dt} = 40\ (\text{cm}^3/\text{s})$	B1
	$A = 4\pi r^2 \quad \frac{dA}{dr} = 8\pi r$	M1A1
	$V = \frac{4}{3}\pi r^3 \quad \frac{dV}{dr} = 4\pi r^2$	M1A1
	$\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dV} \times \frac{dV}{dt}, = 8\pi r \times \frac{1}{4\pi r^2} \times 40 \quad (= \frac{80}{r})$	M1,A1ft
	$r = 4$ so $\frac{80}{4} = 20\ (\text{cm}^2/\text{s})$	dM1A1cao **[9]**

	Any letters can be used for volume and area, inc *SA* for area, but their choice must be used consistently.
B1	State or use $\frac{dV}{dt} = 40\ (\text{cm}^3/\text{s})$ (units not needed)
M1	Attempt to differentiate $4\pi r^2$ with respect to r (Formula for area of sphere is given on formula page)
A1	Correct derivative = d*A*/d*t*
M1	Attempt to differentiate $\frac{4}{3}\pi r^3$ with respect to r (Formula for volume of sphere is given on formula page)
A1	Correct derivative = d*A*/d*t*
M1	Show (or use) a useful chain rule. Terms can be in any order as long as it is possible to obtain d*A*/d*t* from it. OR Use chain rule twice to obtain an expression from which d*A*/d*t* could be obtained.
A1ft	Substitute their expressions for the 3 derivatives in their chain rule. Need not be simplified.
dM1	Use the resulting expression(s) with $r = 4$ to obtain a value for d*A*/d*t* All previous M marks needed.
A1cao +cso	Correct value, units may be missing. Solution must be correct.

Type MS · Connected rates: linking two changing quantities

Answer 6 of 7 · Question 10 part whole · 9 marks

2020 JAN · Paper 02 · extract 2 of 2 · find part whole in this mark scheme extract

Question Number	Scheme	Marks

Type MS · Connected rates: linking two changing quantities

Answer 7 of 7 · Question 7 part b · 5 marks

2020 NOV · Paper 01 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

Number		
7 (a)	$(V=)x^3 \quad (\frac{dV}{dx}=)3x^2 \quad (\text{at } x=2 \quad \frac{dV}{dx}=12)$	M1
	$\frac{dx}{dt}=0.1$	B1 (A1 on ePen)
	$\frac{dV}{dt}=\frac{dV}{dx}\times\frac{dx}{dt}="12"\times 0.1$ oe	M1
	1.2 m^3/s cao oe	A1cao (4)
(b)	(Surface Area =) $6x^2 \quad (\frac{dA}{dx})=12x \quad$ at $x=6$	M1
	$\frac{dA}{dx}=72$	A1
	$\frac{dV}{dx}=108 \quad \frac{dA}{dt}=0.05$	B1
	$\frac{dV}{dt}=\frac{dV}{dx}\times\frac{dx}{dA}\times\frac{dA}{dt}="108"\times"\frac{1}{72}"\times 0.05$	M1
	0.075 m^3/s	A1 (5)
	ALT	
	A = $6x^2$ leading to an expression in A for $V \quad V=\left(\frac{A}{6}\right)^{\frac{3}{2}}$	M1
	$\frac{dV}{dA}=\frac{1}{4}\left(\frac{A}{6}\right)^{\frac{1}{2}}$ oe	A1
	$\frac{dV}{dA}=\frac{3}{2}$ and $\frac{dA}{dt}=0.05$ oe	B1
	$\frac{dV}{dt}=\frac{dV}{dA}\times\frac{dA}{dt}="\frac{3}{2}"\times 0.05$	M1
	1.2 m^3/s cao oe	A1cao [5] **[9]**

Type MS · Connected rates: linking two changing quantities

Answer 7 of 7 · Question 7 part b · 5 marks

2020 NOV · Paper 01 (regional R) · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	Correct expression for Volume, attempt at differentiation to ax^2, a is an integer, $a > 1$.
	B1	$\frac{dx}{dt} = 0.1$, can be explicit or implicitly used in a chain rule.
	M1	For **any correct chain rule**, that would lead to a value for $\frac{dV}{dt}$ and substitution of 0.1 and their value for $\frac{dV}{dx}$. They must show an attempt to find $\frac{dV}{dx}$, need not be a correct attempt, this isn't a dependent mark.
	A1	cao oe
(b)	M1	Correct expression for Surface Area, attempt at differentiation to bx, b is an integer, $b > 1$.
	A1	$\frac{dA}{dx} = 72$
	B1	$\frac{dV}{dx} = 108$ & $\frac{dA}{dt} = 0.05$ clearly stated, implicitly in chain rule or explicitly
	M1	For **any correct chain rule**, that would lead to a value for $\frac{dV}{dt}$ and substitution of 0.05 and their values for $\frac{dA}{dx}$ and $\frac{dV}{dx}$ They must show an attempt to find $\frac{dV}{dx}$, need not be a correct attempt, this isn't a dependent mark.
	A1	cao oe
(b) ALT	M1	Correct expression for area, attempt to rearrange, expression for V in terms of A
	A1	oe
	B1	Both derivatives clearly stated, implicitly in a chain rule or explicitly
	M1	For **any correct chain rule**, that would lead to a value for $\frac{dV}{dt}$ and substitution of 0.05 and their value for $\frac{dV}{dA}$. $\frac{dV}{dA}$ doesn't need to come from correct working, but there must have been some attempt to find an expression for V in terms of A and $\frac{dV}{dA}$ presented somewhere in the working.
	A1	cao oe

Type MS · Connected rates: linking two changing quantities

Answer 7 of 7 · Question 7 part b · 5 marks

2020 NOV · Paper 01 (regional R) · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks